

PHANTOM TWIN

*Behavioral Intelligence Meets Active Deception,
Github: <https://github.com/Boredem125/Phantom-Twin>, Hirepro isn't taking the code*

AI-Powered Behavioral Anomaly Detection with Active Deception

THEME

Cybersecurity

CATEGORY

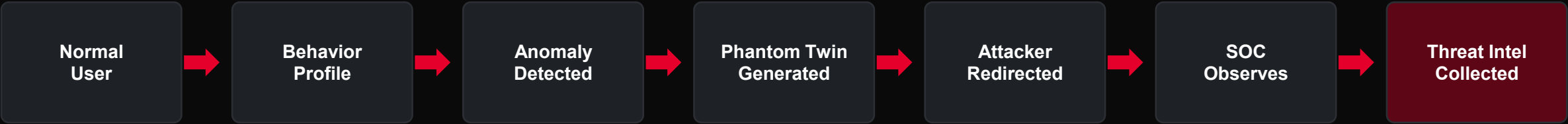
Software

PROBLEM STATEMENT

AI-Powered Behavioral Anomaly Detection

Proposed Solution

Instead of stopping at anomaly detection, Phantom Twin actively deceives the attacker.



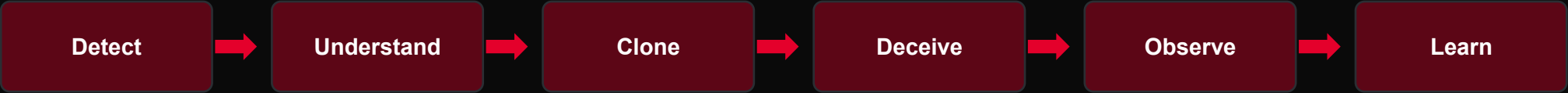
INNOVATION

Detection is a starting point, not the outcome.

TRADITIONAL
SOC



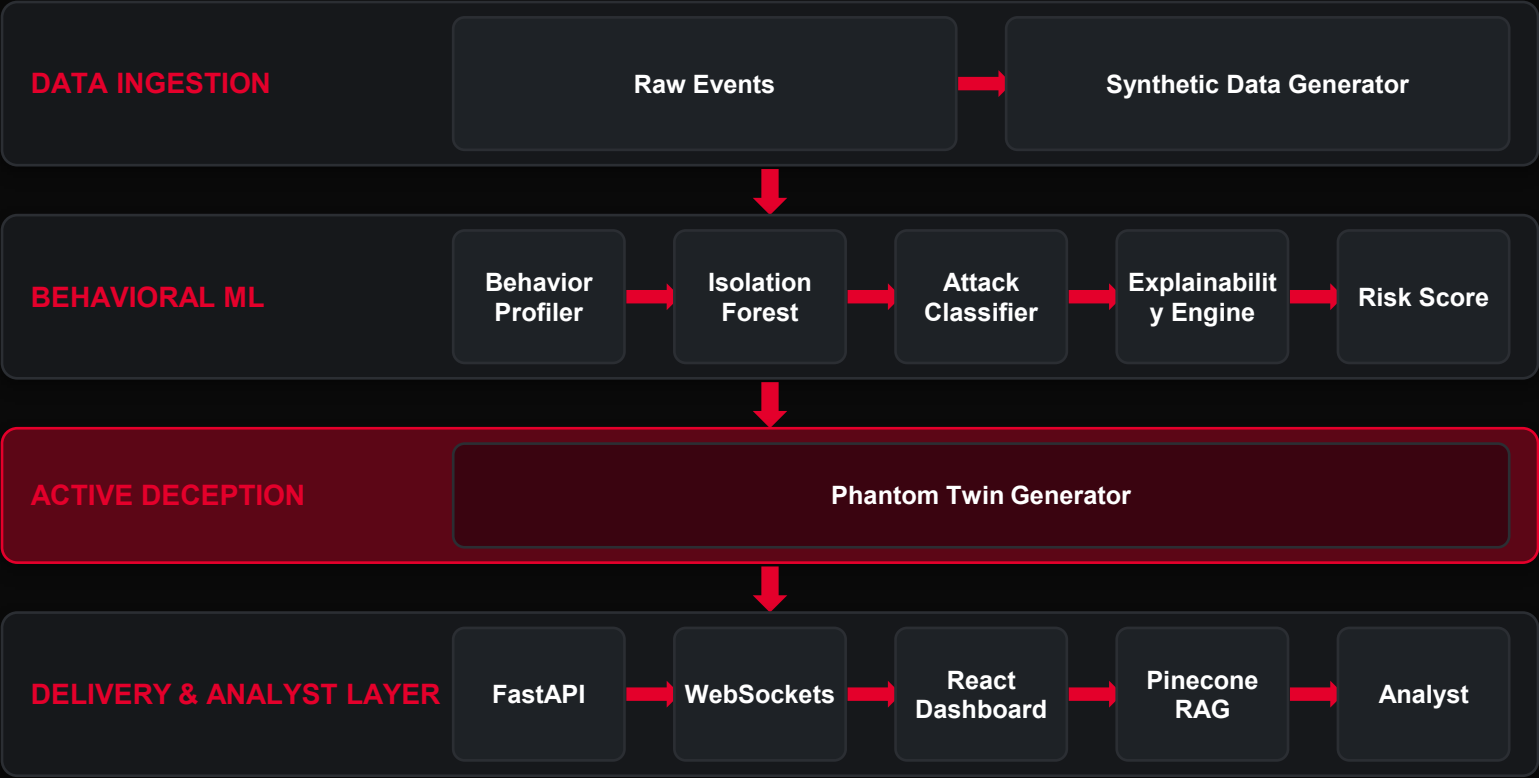
PHANTOM
TWIN



*Detection is reactive. **Deception is proactive.***

TECHNICAL APPROACH

System Architecture



TECHNOLOGY STACK

- **Backend**
Python, FastAPI
- **Frontend**
React, Tailwind
- **Machine Learning**
Isolation Forest, NumPy, Pandas, Faker
- **Explainability**
LLM AI Assistant
- **Vector Database**
Pinecone

Built to Scale, Designed to Survive Real Conditions

0.986

BINARY AUC-ROC

90.2%

PRECISION @ TOP-1% ALERTS

0.10%

FALSE POSITIVE RATE

FEASIBILITY

- ✓ Enterprise scalable design
- ✓ Real-time event streaming
- ✓ Covers users and OT devices
- ✓ Synthetic dataset generation

CHALLENGE

Cold start



Concept drift



Class imbalance



MITIGATION

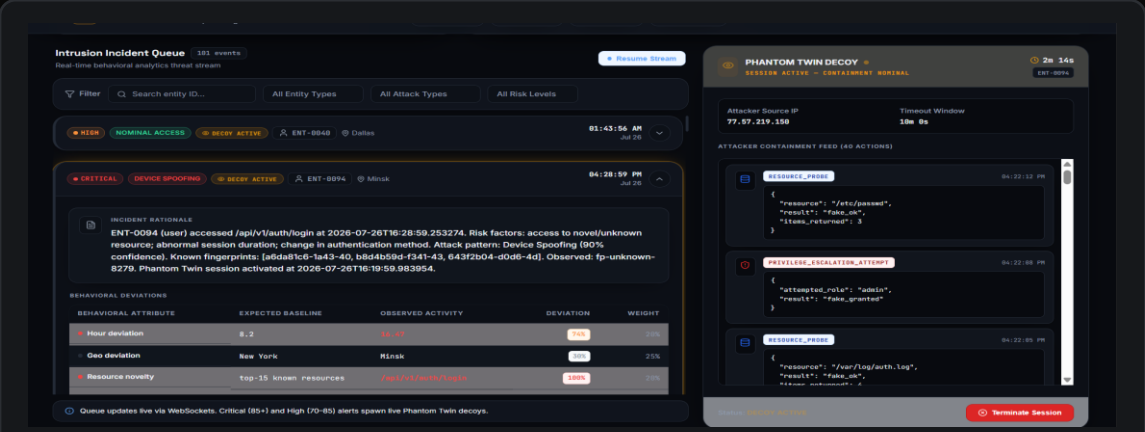
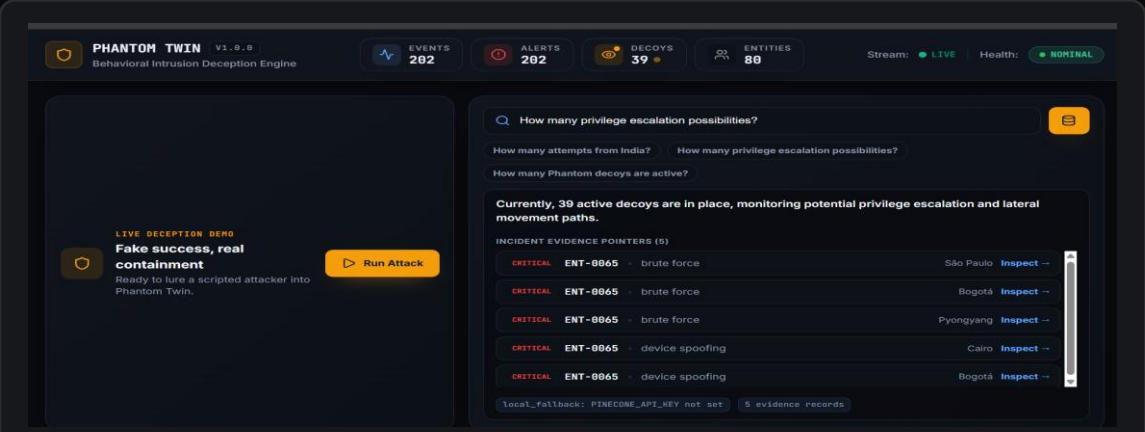
Peer profile bootstrapping

Continuous learning loop

Hybrid detection: Isolation Forest + stateful rules

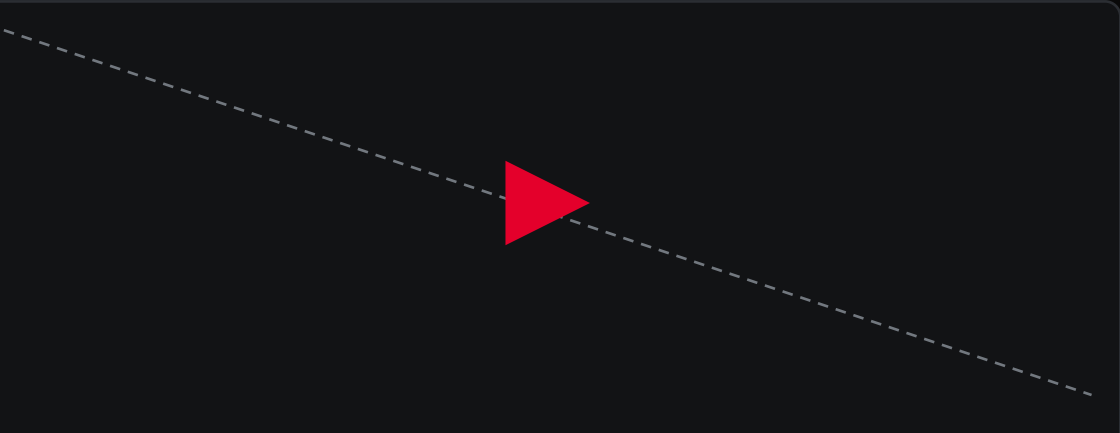
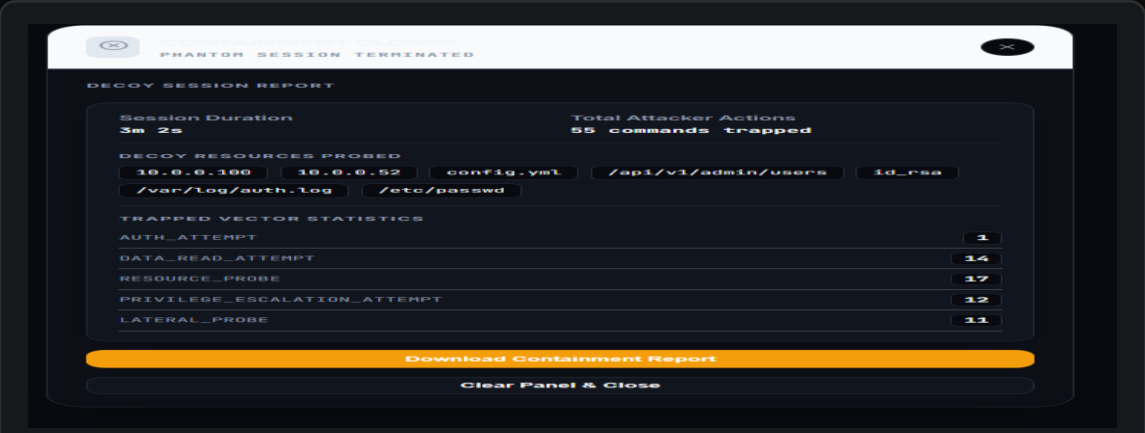
Also delivered: Natural Language Analyst Assistant · Downloadable Containment Reports

Platform in Action



Live Threat Dashboard & Containment Feed

Every attacker action is captured live after silent redirection into the Phantom Twin.



DEMO VIDEO

Please Ctrl+Click on the play button

Phantom Session Report

Downloadable forensic report: attacker actions, resources probed, containment duration.

Grounded in Established Security Research

REFERENCES

- Honeywell Problem Statement
- Behavioral Anomaly Detection literature
- Isolation Forest (Liu et al.)
- MITRE ATT&CK Framework
- Explainable AI (XAI) research
- Pinecone vector database docs
- FastAPI documentation
- React documentation

SOLUTION COVERAGE

- | | |
|---------------------------------|----------------------------|
| ✓ Behavioral Profiling | ✓ Sequence-aware Detection |
| ✓ Attack Classification | ✓ Explainability |
| ✓ Analyst Dashboard | ✓ Cold-start Handling |
| ✓ Concept Drift Handling | ✓ Real-time Streaming |
| ✓ AI Analyst Assistant | ✓ Active Deception Layer |
| ✓ Downloadable Incident Reports | |

PHANTOM TWIN

We don't just detect attackers. We let them expose themselves.